Solar Powered Golf Cart System for On-Campus University Use

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Abstract- **The aim of this Capstone project, considering the Kingdom's 2030 vision as well as University's mission, is to propose and design a sustainable solution for charging on-campus golf carts by solar energy. The design constitutes primarily of two stages. First is to install solar panel on the electric vehicle's roof to augment the traditional batteries and thus, extend the battery run time. Second stage is to build a grid-connected solar charging station where on campus EVs can be parked on rote basis and take advantage of the high solar irradiation while in parking mode. This in turn will help reduce in-house electricity consumption caused by the conventional charging method. The project upon successful completion is expected to serve as a teaching tool for Effat University's students. Moreover, it will serve as a demonstration with the potential to be later adopted by the broader community in hospitality and other sprawling campuses.**

Keywords- Golf cart, solar power, battery charging.

I. INTRODUCTION

Role of energy is indispensable in everyday life. Renewable energy, such as, waves, sun, tidal, solar and geothermal, is a naturally abundant resource with a great potential to provide sustainable and environmentally friendly energy solutions. Renewable resources are spatially distributed over large geographical locations, with almost infinite supply. The challenge is to use these sources, despite their intermittent nature and high capital costs, in a most cost-effective way.

The electrical vehicle, commonly known as EV, first appeared in the mid-19th century and was developed further in the beginning of the 21st century. It is an electric car which either runs on a single source of energy or two sources of energy (hybrid-type). Based on how they are powered they can be classified into many types such as Hybrid Electric Vehicles, Plug-in Hybrid Electric Vehicles, Battery Electric Vehicles, etc. EVs are also categorized on their functionality as regular cars and golf carts. Golf carts are a type electrical vehicle, invented in 1932, used as a handy means of transport in large campuses like resorts, universities and public parks. Golf Cart batteries are conventionally powered by electricity. Some golf carts are fueled by gasoline. The golf-cart is generally in high demand due to its robust design, convenience, and versatility. It comes in handy in large, spread-out campuses, such as universities, hospitals, airports, golf courses, recreational parks/ resorts, etc. The solar-powered golf carts are an alternative to gas or regular charged electric golf cart, and it is also becoming increasingly popular [7].

Several previous applications of solar powered golf carts were reviewed to gain further understanding about their methodologies, design, and ideas. In Addition, the EV Solar station history for the most popular solar stations since (2014).The first application of solar-powered golf carts can be dated to 2009 at TEMPLE University, Philadelphia [1]. The student's idea was partially based off the suggestion made by their previous project: the hybrid electric vehicle. They installed a regenerative braking system, AC power cord, and an automatic switch from gas to electric power. The student team concluded that installing an AC power cord and an automatic switch from gas to electric were both feasible and significant to the project. And for further improving the efficiency of the golf cart they decided to first, install a solar panel then, to make the vehicle purely electric. Also, to increase the energy storage capacity: the I.C. engine was removed to create more space for additional energy storage devices. In 2010 at Bathurst Glen, Richmond Hill, Ontario [2] a Golf course facility, did an evaluation report of solar assisted electric and gas golf carts. They developed a green-powered golf cart, to reduce air pollution from gas-powered golf carts. The time needed to refuel the golf cart by gas can take few minutes unlike electricity, However, due to its other drawbacks electric carts are advantageous over gasoline ones. So, they developed golf carts generated its electricity from the sun, which has improvements in both cost and performance. And further benefits include saving money, extending the daily time service, reducing the need for stand-by carts, and reducing charging costs. In 2011 at The University of Central Florida, Orlando, Florida [3], Senior students implemented and designed a more energy efficient golf cart that changes its energy consumption based on the driver's needs. One of the design issues they faced was to use the batteries in a more efficient manner. They fixed that issue by discovering that the golf cart has the capability to switch between three modes of operation: efficient, high performance and power saving. Otherwise, the typical mode of operation is standard mode. In 2015 at Tennessee Tech University, Cookeville [4], four manufacturing and

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engineering technology seniors spent the semester building a stronger roof and rewiring the entire car, so it could draw enough power to run for hours. Cumberland Auto donated the work for the cart's paint job and some of the sandblasting. The team mounted a 60-pound solar panel on the roof and attached it to six batteries, which would give it enough power to keep it running for hours or even days in the right weather. From the same university which is the Last application was in 2017 at University of Tennessee, Knoxville [5] by Honored project students called semiautonomous golf cart Their cart has some semi-autonomous features designed into the cart, such as the ability to stop when obstructed, and steer around moderate obstructions toward a selected GPS target location, with manual oversight and override capabilities for safety.

Saudi Arabia by far has been a fossil fuel-based economy. However, in accordance with the Saudi Arabia government, the Vision 2030 is targeting to make energy production and consumption more sustainable and diverse. There is a high emphasis on the use of renewable energy sources. Solar potential in Saudi Arabia saw interest in solar energy applications as early as 1980s and with the passage of time, it is only increasing in scope. There is an agreement for the Kingdom of Saudi Arabia to implement solar energy plan for 2030, which is the largest in the world in the field of solar energy production. The agreement sets the basis for the development of Saudi Arabia's solar energy sector, under which a new solar power plant will be built, and solar panels will be manufactured and developed in Saudi Arabia to generate 150 and 200 GW by 2030, worth 200 billion dollars. It is expected that this agreement and the projects that will result from it will help Saudi Arabia in providing the oil. Also, an additional 100 thousand jobs will be created in Saudi Arabia [6].

The mobility of EVs has become an essential part of the energy transition and will imply significant changes for vehicle manufacturers, governments, companies, and individuals. With the new technologies, the electric vehicle industry is growing rapidly. This has led to electric vehicles which can go further per charge, for less money than traditional fuel powered vehicles. Conventional EVs use electricity from fossil fuels, adding to carbon footprint. Solar powered cars have the ability to eliminate this problem and provide a more sustainable solution.

Effat University's on-campus golf carts demand is significant. Each golf cart is charged at least 5 hours daily to be ready for next day using conventional method. Instead, we can harvest the wasted solar energy by using it to power these Golf Carts. This system will help to extend the battery life and reduce electricity consumption. Additionally, it will help to position Effat University as a strong advocate of clean energy drive as per Saudi 2030 vision.

II. DESIGN

The design of solar-powered golf carts system consists of two stages. First stage of the project is to install a solar panel on one of the golf cart's rooftop to augment charging by electrical means, as depicted in Fig. 1. Second stage is to design a solar charging station compatible with all of the university's golf carts, as depicted in Fig. 2.The system's main components will be namely, the existing golf cart batteries, solar panels, charging controller. Figure 3 shows the design strategy for parking slot.

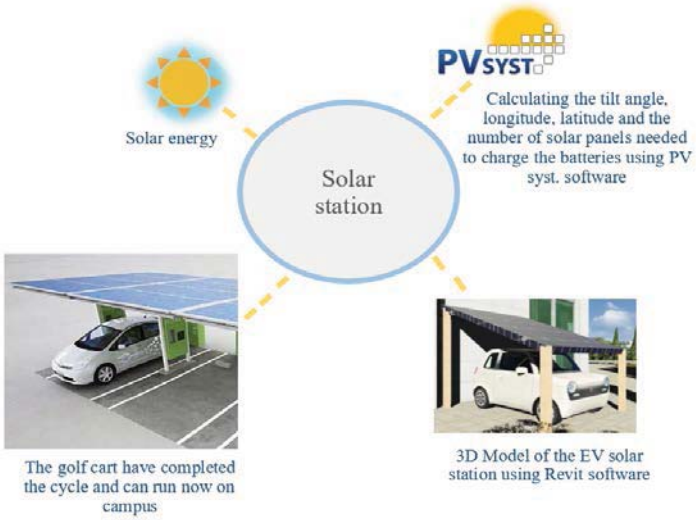

Fig. 1 System Architecture for the golf cart solar statoin

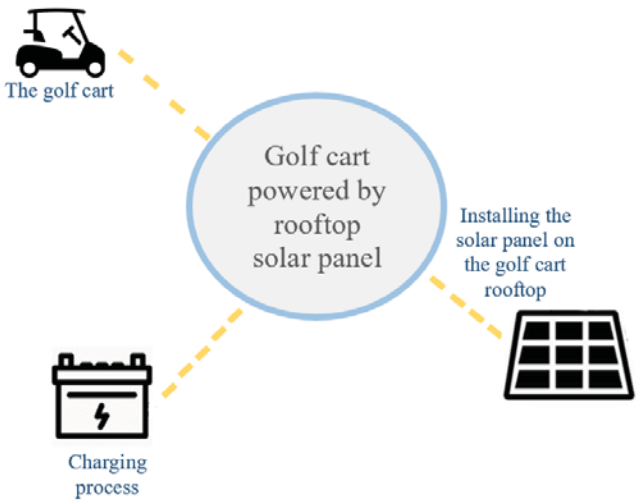

Fig. 2 System Architecture for Rooftop solar panel

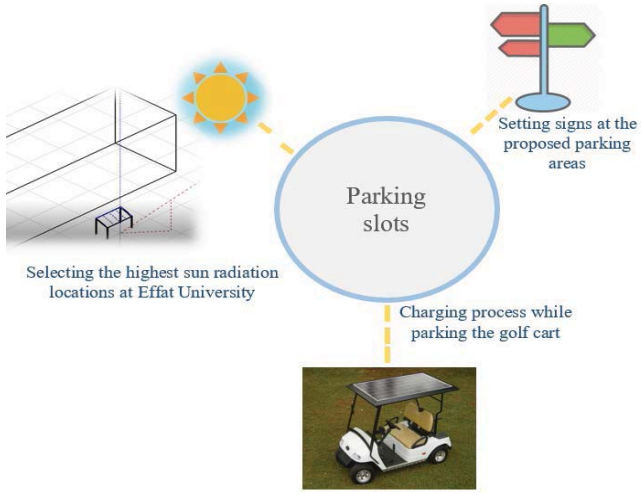

Fig. 3 System Architecture for Parking Slots

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III. METHODOLOGY

The project methodology is a sequence of the most important things to the less important. First higher priority for this project were the location, if we got the best location which has the best sun angle with no shaded area or with a minimum shade area this will be the best location for the golf cart to be parked there. It was found as a result of siting analyses that the space in front of Effat library and cultural museum would be most suited location, since the area is almost 90% shade free throughout the year. A parking sign for the golf cart will be erected in the proposed location.

Each level will need a calculation process using the PV SYST software to figure out the panel's specifications and its extra details.

The electric vehicle's three main components are the electric motor, the control unit, and the battery (Fig. 4). The controller takes power from the battery and passes it to the electric motor. Before passing the current to the engine, the controller converts the DC current to AC.

The electric motor then converts the electric energy into mechanical energy. The mechanical energy moves the car forward, and the control devices act as a medium between the battery and the engine. It transfers the energy from the battery and pumps it to the engine to operate. The variable voltmeters are located between the engine and the controller. It provides the appropriate power for the engine. The engine does not initially output any volt but when pressed fully it gives the maximum output.

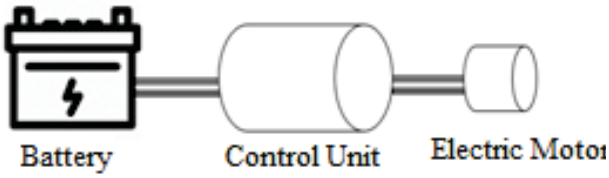

Fig. 4 Electric vehicle components

Tables 1 and 2 show the features and specifications of the components and elements used for implementation and prototype designing in the rooftop golf cart.

Table 1 Project features

Component	Function	Features
The Golf Cart	-Facilitate mobility around campus. -Gives the smoothest ride. - The structure provides safety and support.	- Electrical golf cart from the campus. - White golf cart for four seats.
Golf cart's battery	- A device that produce voltage. - Store the energy needed to move the golf cart.	- Deep-Cycle Flooded/Wet Lead-Acid Battery. - Model: T-105. - The six batteries are connected in series with 36V total load. - Volt/Battery = 6V.
Solar Panel	- Generate electricity through the photovoltaic effect.	- Mono-crystalline. -Power in Watts = 150W. - The Dimensions length 147.8 cm x width 67.8 cm.
Charge Controller	- Prevents Battery over charge. - Prevent battery discharge at night through the solar modules.	- Pulse Width Modulation (PWM) type. - It limits the rate at which electric current is added to or drawn from electric batteries.

Table 2 Project specifications

Equipment	Power	Current	Voltage
Solar Panelsof Mono-crystalline type x3 Modules in series	450 W	37.5 Ah	12 V
Golf cart's Batteries x6	6660 W	185 Ah	36 V
PWM Charge Controller	———	———	12/ 24 V auto recognizer.

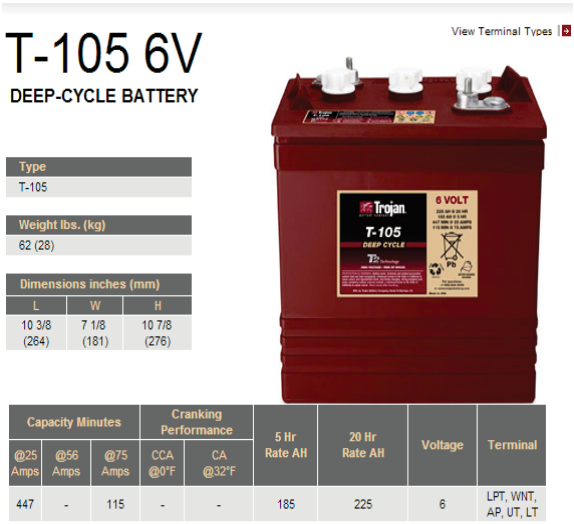

Fig. 5 Golf cart battery's datasheet

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Fig. 6 Solar charge controller's datasheet

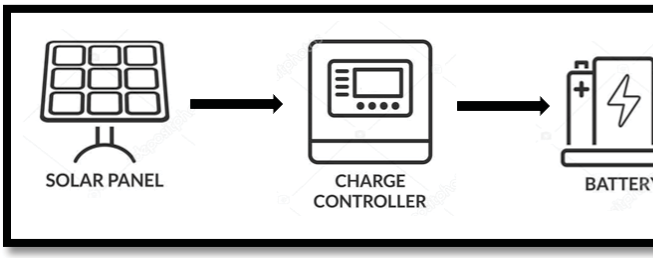

Fig. 7 Charging Process

The battery type used is shown in Fig.5 whereas the charge controller used in shown in Fig. 6. The charging system is as shown in Fig. 7. Mono-crystalline panel, shown in Fig. 8, is the best suitable type for charging EV's batteries because of its high efficiency. To install this system a location study was made to reach the best zero shade location and to have a full sunlight, using PVSYST software to discover the get the best sun position.

Fig. 8 Monocrystalline panel that used in the system

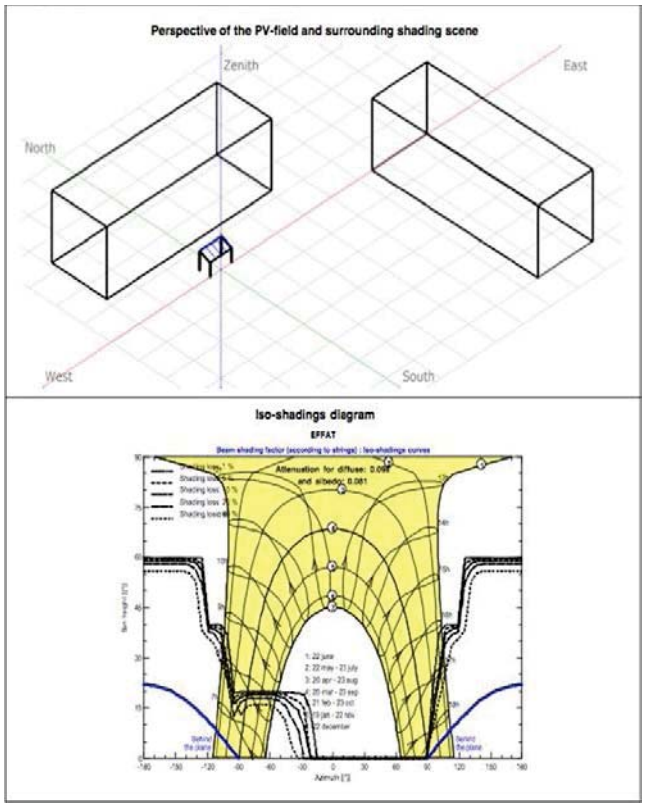

Fig. 9 Location study using PVSYST

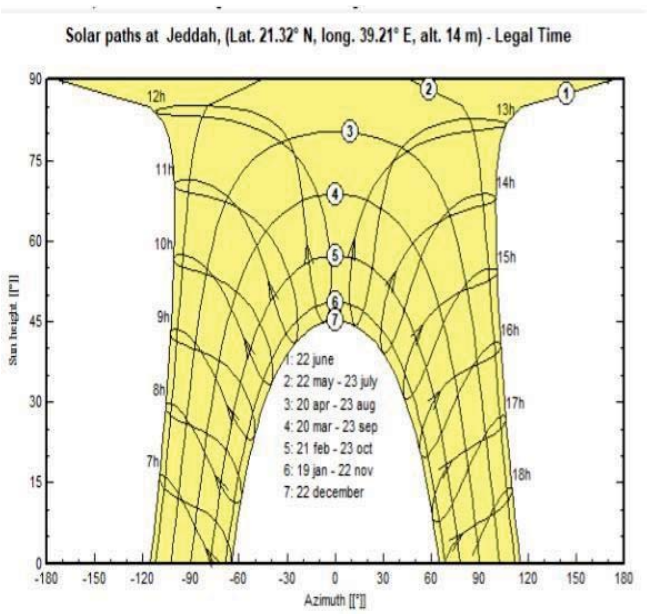

Fig. 10 Sun path diagram using PVSYST

IV. SIMULATION

The system software analysis consist of the location selection, sun path and radiation, solar panels tilt angle and the number of modules needed in the system. Also the numerical simulation and system losses in general, so it contains three solar panels with 150W connected in series to get the maximum voltage including six-existing batteries with 6 volts each with total voltage of 36 volts (Fig. 9-14).

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Grid-Connected System: Simulation parameters

Project : EFFAT

Geographical Site	**Jeddah**	Country	**Saudi Arabia**
Situation	Latitude 21.47° N	Longitude	39.21° E
Time defined as	Legal Time Time zone UT+3	Altitude	22 m
	Albedo 0.20		
Meteo data:	**Jeddah** Meteonorm 7.2 (1998-2002). Sat=0% - Synthetic		

Simulation variant : EFFAT

Simulation date 01/11/18 12h30

Simulation parameters	System type	**Sheds on a building**	
Collector Plane Orientation	Tilt	22°	Azimuth 0°
Models used	Transposition	Perez	Diffuse Perez, Meteonorm
Horizon	Free Horizon		
Near Shadings	According to strings		Electrical effect 100 %

PV Array Characteristics

PV module Si-poly	Model	**STP 320-20/Wem+ (Superpoly)**	
Original PVsyst database	Manufacturer	Suntech	
Number of PV modules	In series	3 modules	In parallel 1 strings
Total number of PV modules	Nb. modules	3	Unit Nom. Power 150 Wp
Array global power	Nominal (STC)	450 Wp	At operating cond. 1146 Wp (50°C)
Array operating characteristics (50°C)	U mpp	132 V	I mpp 8.7 A
Total area	Module area	7.8 m²	Cell area 7.0 m²

Fig. 11 PV Syst numerical simulation

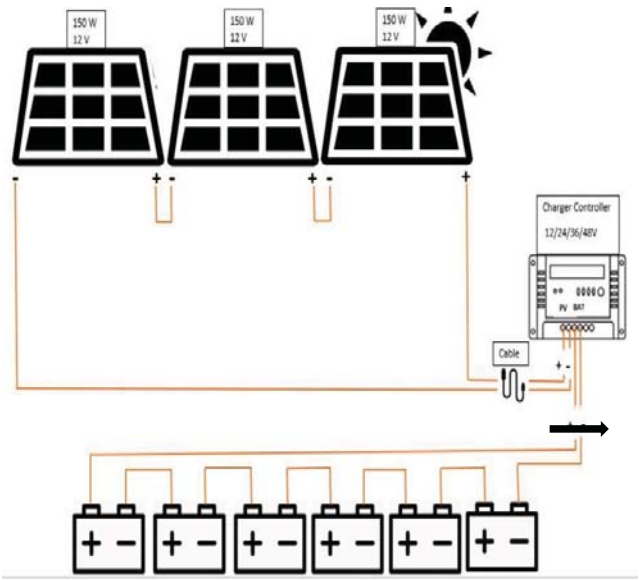

Fig. 12 The system connection

V. RESULTS

The Final analysis, a total of 3x 150W solar panels in this system, was able to charge the battery to work 20 hours in a day. The energy required to fully charged the battery is

- W= Ah x V =185Ah x (6x6) V = 6660 Wh
- Working Hours = 20 hrs.
  Based on the battery datasheet

With 37.5 Ah, 12 V and 450Wh given by the 3 solar panel modules the batteries will successfully be 50% charge for maximum 8hr

W = Ah x V = 37.5Ah x 12 V = 450 Wh

- 50% Charging Hours = 6 -8 hrs.
  W= 450Wh x 8 = 3600 Wh
- 100% Charging Hours = 14-15 hrs.
  W= 450Wh x 15 = 6750 Wh

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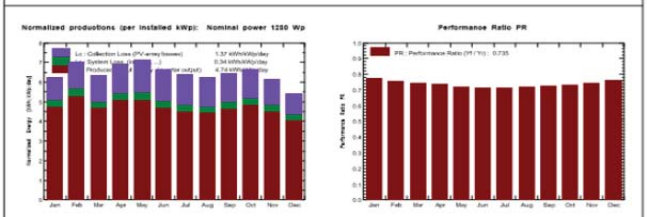
Grid-Connected System: Main results

Project : EFFAT
Simulation variant : EFFAT

Main system parameters	System type	**Sheds on a building**	
Near Shadings	According to strings		Electrical effect 100 %
PV Field Orientation	tilt 22°		azimuth 0°
PV modules	Model	STP 320-20/Wem+ (Superpoly)	150 Wp
PV Array	Nb. of modules	3	Pnom total 450 Wp
Inverter	Model	UNO-DM-1.2-TL-PLUS	Pnom 1200 W ac
User's needs	Unlimited load (grid)		

Main simulation results
System Production — **Produced Energy** **2215 kWh/year** — Specific prod. 1731 kWh/kWp/year
Performance Ratio PR 73.48 %



EFFAT
Balances and main results

	GlobHor kWh/m²	DiffHor kWh/m²	T Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR
January	149.5	37.50	22.06	193.4	184.1	203.6	190.4	0.769
February	162.1	37.33	23.30	198.1	187.0	203.2	189.8	0.756
March	180.7	73.21	25.59	196.0	182.9	199.8	186.6	0.744
April	208.2	69.07	28.50	208.5	196.5	210.0	196.3	0.736
May	237.8	65.83	31.26	220.6	207.8	218.3	203.4	0.720
June	221.2	79.00	32.10	199.2	183.6	194.9	181.1	0.710
July	216.2	89.76	33.65	198.0	182.8	193.5	179.7	0.709
August	198.2	92.55	33.45	193.1	179.7	190.7	177.8	0.719
September	183.9	71.68	31.00	193.3	181.3	192.6	179.9	0.727
October	178.0	55.60	29.43	206.5	194.4	206.0	192.7	0.729
November	145.1	44.81	26.44	183.3	171.8	187.0	175.1	0.746
December	128.9	46.46	23.90	167.5	156.6	174.2	162.7	0.759
Year	2209.9	762.80	28.42	2355.3	2208.3	2373.7	2215.4	0.735

Legends: GlobHor Horizontal global irradiation; DiffHor Horizontal diffuse irradiation; T Amb Ambient Temperature; GlobInc Global incident in coll. plane; GlobEff Effective Global, corr. for IAM and shadings; EArray Effective energy at the output of the array; E_Grid Energy injected into grid; PR Performance Ratio

Fig. 13 System simulation

VI. CONCLUSION

Energy security, limited fossil fuel reserves and the associated environmental concerns are among key drivers pushing Saudi Arabia towards alternative forms of energy solar energy is one of the most important renewable technologies that have a great potential due to the enormous energy that the sun emits year-round. The kingdom of Saudi Arabia has a huge potential of using solar energy but they are not as actively investing in it. From this perspective, we suggest incremental, small-scale solar energy projects for public awareness and for wider adoption. To that end, we have designed a solar-powered golf carts system for on-campus use to reduce grid electricity consumption, get rid of the dependency on fossil fuels, reduce the greenhouse gas emissions and thus, pave way for green and efficient university environment.

Three mono-crystalline type panels were used in this project to charge 50% of the golf cart batteries for 6-8 hours, and 14- 15 hours to be 100% charged. An initial cost of 300 SR per year of the annual savings will be saved on electricity by solar charging with only one panel on golf cart, with 16 years simple payback period for both educational and residential sectors with 0.18 SR/ kWh tariff. If solar charging station is implemented whereby golf carts can be charged in turns, this will reduce the payback period significantly, and further increase the savings on electricity consumption.

The prototype took two working days to be implemented and six months of designing and choosing the

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most correct methods. The prototype is 100% ready to charge the golf-cart batteries from the solar panels. The project was tested before with a tilt angle then the solar panels were installed on the golf cart rooftop. Some of the challenges in proposing solar hybrid Electric Vehicles (EVs) are the high initial investment, expensive batteries, and limited variety of vehicles. Most batteries have to be replaced after every 5 years thus increasing the operational costs for EV system.

Fig. 14 Suggested parking slots after location study

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